Technical Progress Report

Mode-II Reference Verification, Adaptive Remeshing, State Transfer, and Parallelization Assessment for Phase-Field Fracture in Abaqus

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Objective Accurate phase-field fracture with locally refined meshes, controlled state transfer, and a defensible computational-cost reduction.

Progress since the 24 July report. The work has advanced to a Mode-II verification and adaptive-production branch: corrected coarse/intermediate/fine uniform references (H0/H1/H2), two endpoint-complete locally refined calculations, and an executable-path assessment of nonmatching state transfer.

Major supported result. The fine H2 reference calculation was extended to characterize the near-peak response more completely. It remained identical to the previous H2 response over the shared loading range, developed an interior reaction-force peak of 0.3581 kN at $u_1 = 0.00950$ mm, and subsequently terminated by nonlinear solver divergence at $u_1 = 0.0097625$ mm. H1 and H2 closely agree before and near the peak; complete post-peak convergence remains unresolved.

Current evidence boundary. Crack-path convergence fails its frozen matched-state gate. Runtime state ingestion fails. A second evolving remesh, general thread safety, MPI/hybrid safety, and an online adaptive-remesh loop are not established.

Pre-peak uniform verification	PASS	Adaptive MM/PK5 production	PASS
Adaptive-to-uniform accuracy	HOLD	Matched-state crack path	FAIL
Nonmatching transfer artifact	SUPPORTED	Mechanical re-equilibration claim	FAIL
Runtime state ingestion	FAIL	Second evolving remesh	HOLD
Parallel thread safety	UNRESOLVED	MPI / hybrid safety	UNRESOLVED
Online adaptive remeshing	NOT YET CLAIMED	Thesis integration	SUPPORTED

Adaptive comparison boundary. The two locally refined solutions agree closely with each other, but their late-loading reaction-force magnitude is approximately half that of the uniform references. The deck and extraction audit found no justified constant normalization factor. Adaptive-to-uniform accuracy and accuracy-versus-cost are therefore held pending curve-level reconciliation.

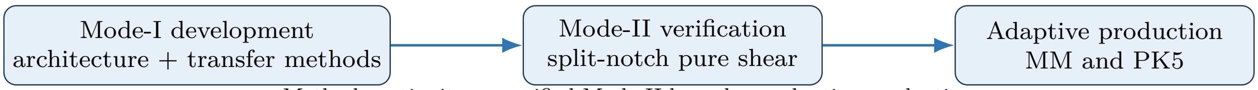
Evidence snapshot: `0b36c8ed3e5420f34ed66f6468c5994ad3f3e0f4`.

1. From Mode I development to Mode II verification

The 24 July report documented the Mode-I development benchmark, layered UEL/UMAT/UEXTERNALDB architecture, MISESERI-guided offline refinement, bounded transfer, fixed-active-set failure, and offline KKT correction. None of that evidence is invalidated. The present Mode-II branch is a later, more demanding verification and adaptive-production benchmark that retains the same methodology: one physical mesh reconstructed as coincident phase, mechanics, and visualization layers; deterministic packages; and claim-by-claim validation.

The Mode-II specimen uses split notch faces and horizontal shear loading. The qualified FRACFIX formulation uses $l_0 = 0.015$ mm, $G_c = 0.0027$ kN/mm, $E = 210$ kN/mm^2, $\nu = 0.3$, unit thickness, and residual stiffness $k = 10^{-7}$. Four-node and three-node elements are supported in three coincident logical layers. U1 is the quadrilateral phase-field UEL, U2 the quadrilateral displacement UEL, U3 the triangular phase-field UEL, and U4 the triangular displacement UEL. Separate passive CPE4/CPE3 facsimile elements form the output layer.

The current corrected phase-field UEL formulation (internally FRACFIX) uses the following producer contract: SDV14 carries the phase used by the mechanical U2/U4 layer; SDV15 is the current solved phase from U1/U3; and SDV16 is history H from the mechanical/history path. SDV14 and SDV15 may differ during staggered execution. N_{phys} denotes the number of physical elements.



Method continuity → verified Mode-II branch → adaptive production.

Reason for the transition. Mode II supplies the current common reference/adaptive comparison problem and exercises mixed-mode-sensitive localization. It is not a replacement for the Mode-I development evidence; it is the next verification and production branch.

2. Uniform H0/H1/H2 reference verification

Metric	H0	H1	H2	Interpretation
Physical elements	3,930	12,064	33,852	$h/l_0 = 2.0, 1.0, 0.5$
Valid u_1 maximum (mm)	0.010000	0.0096325	0.0097625	H1/H2 common: 0–0.0096325
K_0 (kN/mm)	46.1396	45.9035	45.8741	H2–H1: -0.064%
$u_1(d \geq 0.5)$ (mm)	0.008250	0.007750	0.007750	H1/H2 parity
H2 vs H1 L_2 / work area	–	–	0.536% / 0.287%	new common domain
H2−H1 matched RF / $d_{\max}$	–	–	−1.445% / +0.001948	at $u_{\max}^{\text{common}}$
CPU / wall (s)	2000/2004	5434/5453	15949/15990	measured serial cost
Termination	endpoint	divergence	divergence	H1/H2 incomplete

H0 reaches the prescribed endpoint. H1 terminates by nonlinear divergence at $u_1 \simeq 0.0096325$ mm. The extended H2 calculation reproduces the previous H2 RF–U and $d_{\max}$ histories exactly through $u_1 = 0.009250$ mm, establishing numerical reproducibility on the overlapping interval. H2 then develops an interior peak $RF_1 = 0.358134$ kN at $u_1 = 0.009500$ mm and terminates by nonlinear divergence at $u_1 = 0.0097625$ mm after Step 2 increment 1906.

The enlarged H1/H2 common domain ends at $u_1 = 0.0096325$ mm. Over it, normalized RF $L_2 = 0.536\%$, absolute work difference = 0.287%, matched RF difference = −1.445%, and matched $d_{\max}$ difference = +0.001948. Origin-OLS stiffness differs by only −0.064%. Both reach $d_{\max} \geq 0.5$ at 0.00775 mm, whereas $d_{\max} \geq 0.9$ occurs at 0.00850 mm for H1 and 0.00800 mm for H2. Thus the pre-peak/near-peak global force response is highly consistent, while later localization timing is more mesh-sensitive.

At matched $u_1 = 0.009250$ mm the H1/H2 crack sets ($d \geq 0.5$) have Hausdorff distance 0.005443 mm, above the frozen 0.003750 mm gate. No H2 detailed field frame exists exactly at the new common endpoint; the nearest later H2 frame is approximately 0.0097500 mm and lies outside the H1 domain. Accordingly, no new-endpoint Hausdorff value is claimed. Direct validated phase-field fracture-energy history is unavailable; ALLSE or ALLWK−ALLSE is not substituted, and energy convergence remains unresolved.

pre-peak mesh-refinement consistency	PASS
damage-initiation consistency	PASS
H1 minimum pre-peak/global comparator	SUPPORTED
H2 fine spatial diagnostic	SUPPORTED
matched-state crack-path convergence	FAIL
global peak convergence	UNRESOLVED
full post-peak uniform convergence	UNRESOLVED
complete uniform fracture reference	**NONE**

3. Uniform-reference figures and interpretation

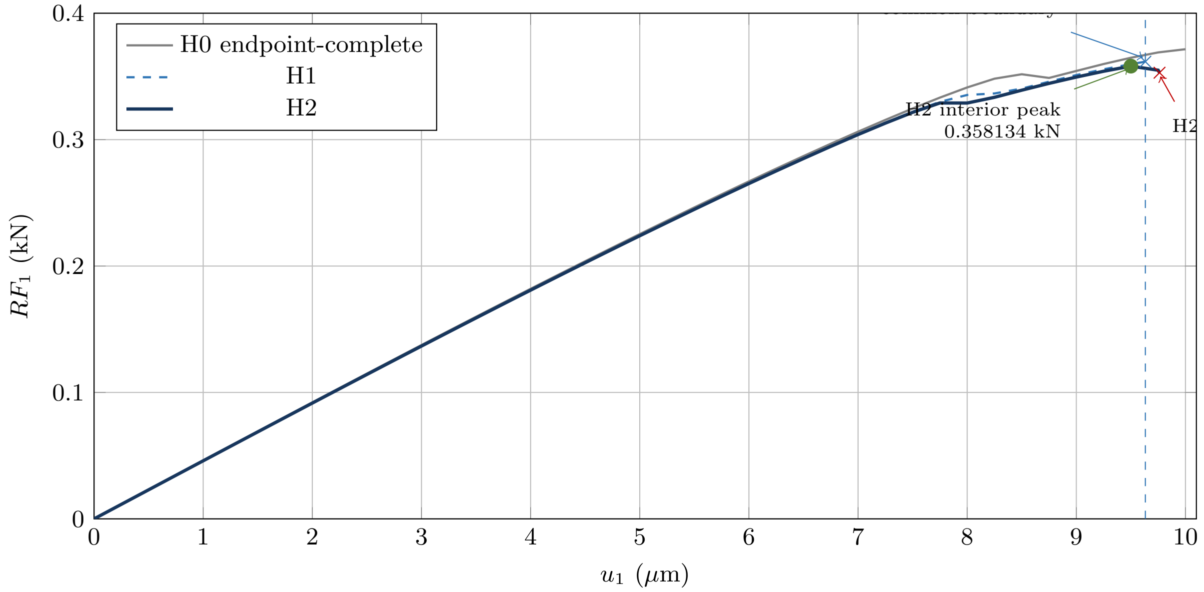
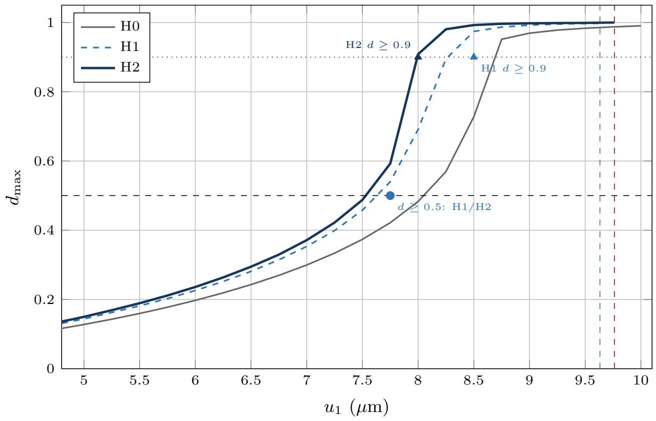
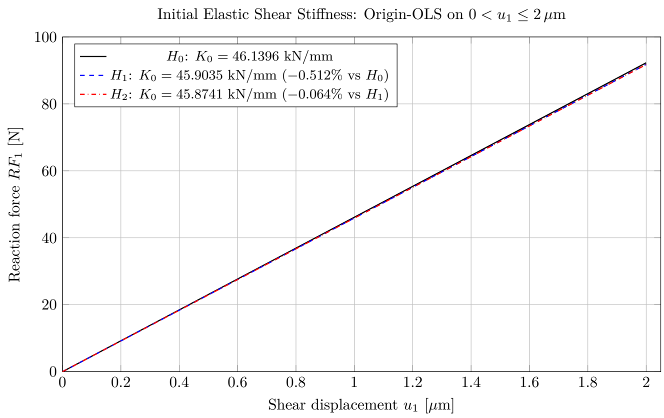


Figure 1. RF_1–u_1 comparison from primary extracted curves. H2 reproduces the previous solution exactly to 9.25 μm, reaches an interior peak at 9.50 μm, and diverges at 9.7625 μm. H1 diverges at 9.6325 μm; global H1/H2 peak convergence remains unresolved.





Figures 2–3. Extended damage evolution and identical origin-OLS stiffness over $0 < u_1 \leq 0.002$ mm. H1/H2 share $d_{\max} \geq 0.5$ initiation at 0.00775 mm; $d_{\max} \geq 0.9$ shifts from 0.00850 mm (H1) to 0.00800 mm (H2). K_0 =46.1396, 45.9035, and 45.8741 kN/mm for H0/H1/H2.

H2's global phase range is 0 to 0.999966 with no overshoot. SDV16 has no negative transitions, but eight small SDV15 decreases occur (worst -3.9768×10^{-4}). History monotonicity is supported; strict solved-phase pointwise irreversibility is not established.

4. Adaptive mesh development and production

MISESERI pre-analysis was used to localize resolution before the fracture solves. Two locally refined meshes were selected for production. The first is a lower-cost, strongly localized mesh with 2,206 physical elements, referred to internally as MM. The second is a finer crack-corridor mesh with 4,894 physical elements, referred to internally as PK5. MM uses minimum/maximum local-size control and has crack-corridor median $h/l_0 = 0.8170$; PK5 uses a 5% error-target criterion and has median $h/l_0 = 0.6033$. They represent different compromises between cost and local fracture-process-zone resolution.

Job	Elements	CPU / wall	Peak RAM / VMEM	Solver result	Scope
Lower-cost mesh (MM) 1386469	2,206	164 / 166 s	4.86 / 7.58 GB	endpoint, 2,500 increments, 0 cutbacks	strong localization
Finer mesh (PK5) 1386470	4,894	366 / 368 s	10.12 / 12.90 GB	endpoint, 2,500 increments, 0 cutbacks	higher corridor resolution

Both jobs completed at $u_1 = 0.010000$ mm with exit status 0. Their mutual maximum relative RF_1 difference is approximately 0.16%, which supports adaptive-to-adaptive repeatability only. A deck/extraction audit confirms that all cases report the same RP RF_1 quantity, sign, units, displacement source, and loading amplitude; nevertheless, the adaptive curves are of order 0.1785 kN near completion while the fine uniform response is of order 0.35 kN near its peak. Matching initial stiffness rules out a constant factor-of-two correction. Until the complete adaptive curves are reconciled against H1, no adaptive-to-uniform accuracy claim is made.

Supported result. Two materially different locally refined meshes completed the prescribed displacement and agree closely with each other.

Held claim. Agreement between the two adaptive meshes alone does not establish agreement with the uniform reference.

5. Measured serial cost and pending adaptive-reference accuracy audit

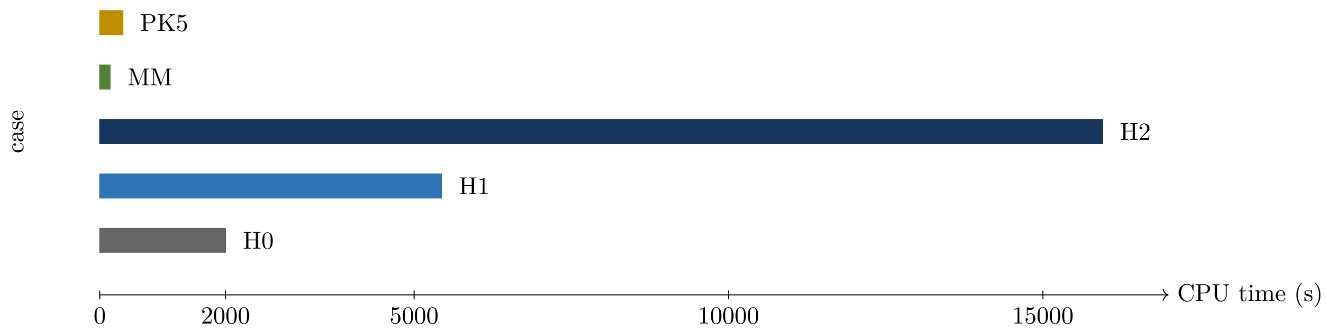


Figure 4. Audited one-CPU time. MM denotes the lower-cost adaptive mesh (2,206 physical elements); PK5 denotes the finer crack-corridor adaptive mesh (4,894 physical elements). H1/H2 are incomplete uniform diagnostics; MM/PK5 are endpoint-complete. Cost alone does not establish adaptive-to-uniform accuracy.

Case	H2 / case CPU ratio	Elements	Scientifically comparable response statement
H0	7.9745×	3,930	coarse uniform baseline; full endpoint
H1	2.9343×	12,064	minimum supported pre-peak comparator; incomplete
H2	1.0000×	33,852	fine spatial diagnostic; numerically incomplete
MM	97.25×	2,206	endpoint-complete; adaptive-reference accuracy HOLD
PK5	43.58×	4,894	endpoint-complete; adaptive-reference accuracy HOLD

H2 consumed 15,949 s CPU and 15,990 s walltime. It used 97.25 times the measured serial CPU time of MM and 43.58 times that of PK5. These are cost ratios, not equivalent-accuracy speedups. The adaptive-to-uniform force comparison remains on HOLD, so an accuracy-versus-cost conclusion is not yet supported.

6. State transfer and the restart claim audit

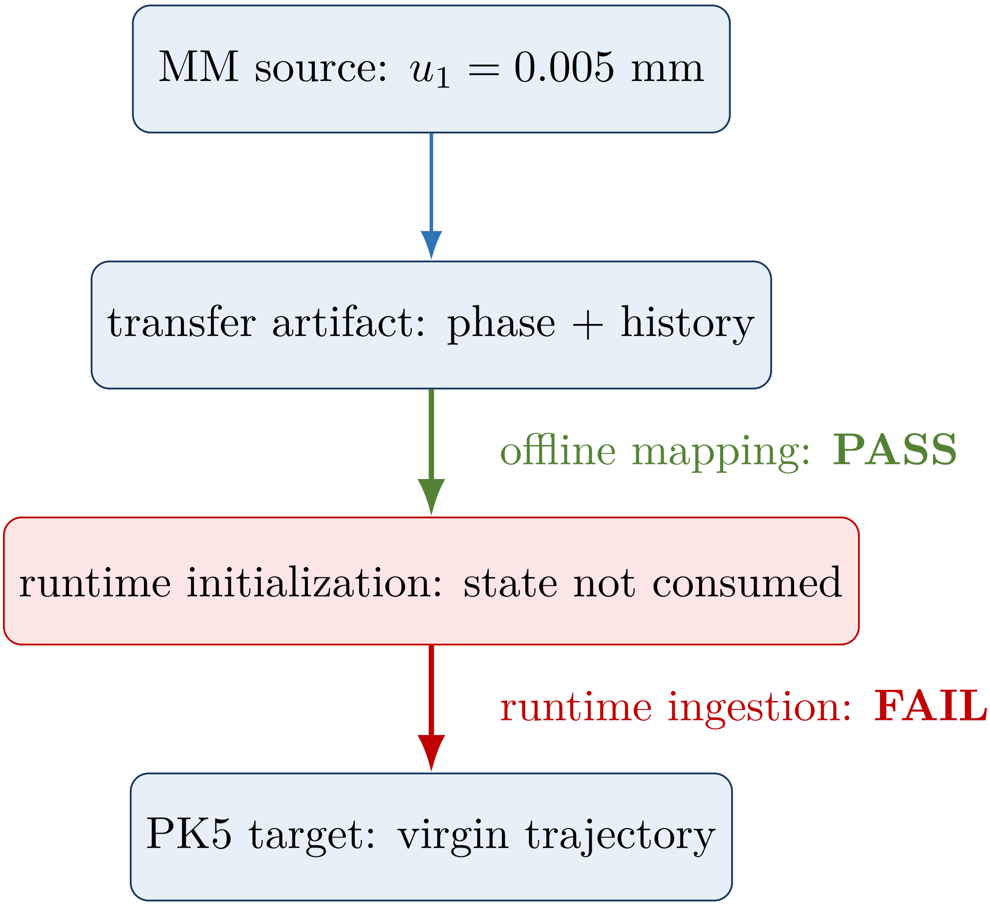


Figure 5. Audited source-to-runtime chain for job 1386471. The artifact exists and offline mapping is traceable, but the executable path does not ingest it. Source: runtime-ingestion audit F43STATE-M2-RUNTIME-INGESTION-AUDIT1.

Job 1386471 used MM Step-1 frame 500 ($u_1 = 0.005000$ mm) as the intended source and a nonmatching PK5 target (4,894 physical; 14,682 layered elements). The artifact contained mapped nodal phase and integration-point history; the offline phase L_2 mapping error was 0.048%, with bounds and coverage checks passing. The target solve then completed to $u_1 = 0.010000$ mm in 2,100 increments, 4,200 equilibrium iterations, and no cutbacks.

Subsequent executable-path verification showed that the prepared transfer artifact was not read by the active solver initialization path: the UEL does not read the passed `SVARS`, mutable `COMMON/KUSER/USRVAR` starts at zero, and nodal phase also starts at zero. Consequently, job 1386471 followed the virgin target-mesh trajectory rather than a transferred-state continuation. Zero RF and ALLSE jumps demonstrate trajectory identity, not transferred-state re-equilibration. Observed phase decreases during the nominal re-equilibration (76 locations, maximum 0.1245) further contradict the intended irreversibility claim.

Current classification. Artifact creation and source-to-artifact trace: SUPPORTED. Artifact-to-running-solver trace: FAIL. Controlled continuation and mechanical re-equilibration claims: FAIL. The second controlled state-transfer/remeshing stage is on HOLD.

7. Parallelization of the ExternalDB / COMMON-block implementation

Supervisor question. Does using ExternalDB and COMMON blocks still prevent parallelization in current Abaqus, and what must change for future parallel applications?

The evidence supports a more precise answer than a blanket prohibition. UEXTERNALDB is a lifecycle callback and does not itself prohibit parallel execution. In pure thread mode, threads share process memory; COMMON blocks, files, and other shared resources require race protection, and Abaqus provides mutex and allocatable-array utilities. MPI ranks are separate processes: ordinary COMMON storage is replicated and process-private, so inter-rank consistency requires explicit ownership, communication, or distribution. Hybrid MPI+threads combines both obligations and requires thread-safe user routines; SIMULIA explicitly states that this precludes COMMON blocks, DATA statements, and SAVE statements. The present shared-state architecture must therefore be refactored, not merely mutex-guarded, before hybrid qualification.[1]

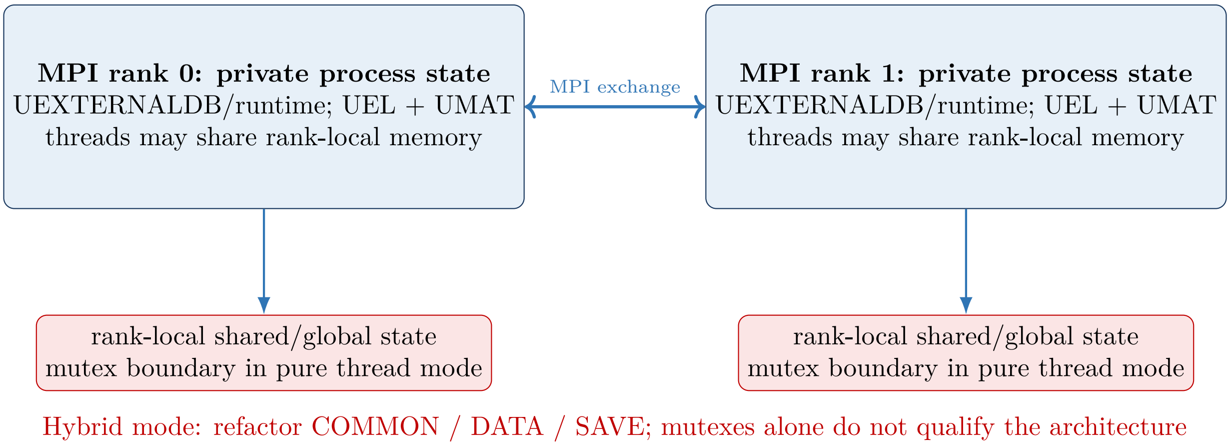


Figure 6. Each MPI rank executes its own UEL/UMAT work with process-private state; threads within a rank share memory. Pure-thread race protection and explicit inter-rank distribution are separate obligations; hybrid execution requires removal/refactoring of COMMON/DATA/SAVE state.

The parallelization qualification study found 38 COMMON, 7 SAVE, 21 DATA, and 87 WRITE occurrences across the audited variants. USRVAR is read/written from UEL and UMAT, and UEXTERNALDB variants perform shared-state loading and file I/O. Static inspection identifies risk; it cannot prove callback order, same-thread ownership, or cross-rank consistency.

[1]SIMULIA, *About User Subroutines and Utilities*, “Parallelization,” and *About Parallel Execution*, “Use with User Subroutines,” Abaqus documentation (accessed 11 August 2026).

8. Parallelization evidence and current status

Execution mode	Abaqus capability / memory model	Current implementation evidence	Status
Serial	ordinary user-subroutine baseline	P3-S failed with signal 11; minimized P3-SM0 and later serial identifier lanes completed; finalized P3-SB evidence retained	SERIAL BASELINE
1 MPI rank, 4 threads	threads share process state	Small repeatability test D2C matched its serial case. Threaded attempt P3-T4 failed compilation before callbacks/solver.	CASE EVIDENCE
Multiple MPI ranks	separate rank address spaces	no proof that COMMON/global state is distributed consistently	UNQUALIFIED
Hybrid MPI + threads	thread-safe routines required; COMMON/DATA/SAVE precluded	no valid scientific hybrid evidence; shared-state refactor required	UNTESTED

A serial diagnostic (P3-S) and minimized callback tests (P3-SM0/P3-SM1) established usable serial identifier-call routes. The later threaded qualification attempt P3-T4 (job 1378242) exited during Intel compilation before callbacks or solver execution. It therefore supplies no threaded scientific evidence.

Positive evidence. The small one-rank/four-thread repeatability test (D2C) shows equality for that case only.

Claim boundary. Abaqus parallel modes being available does not mean this mutable COMMON/SAVE implementation is safe in them. General thread safety, MPI state consistency, and hybrid safety remain separate, unqualified claims.

9. Proposed parallelization subproblem

The next investigation should proceed as a gated engineering subproblem, with no promise of production parallel speedup before numerical equality and determinism are demonstrated.

Stage	Action	Measurable output / pass condition
P-A	Static shared-state and file-I/O inventory	owner, writer, lifecycle, rank/thread scope for every mutable value
P-B	Frozen serial deterministic baseline	repeat RF–U, SDV14/15/16, phase, history ordering, energy and hashes
P-C	One-rank multithread characterization	numerical equality, repeated-run determinism, callback/access trace
P-D	Thread-safe refactor	remove or synchronize unsafe mutable writes; repeat P-B/P-C gates
P-E	Multi-rank distribution design	partition/replicate state explicitly; prove rank-complete ordering and equality
P-F	Hybrid test	equality plus deterministic runtime, speedup and memory scaling

For every stage, compare the complete RF–U curve, SDV14/15/16 fields, history ordering, phase irreversibility, energies, deterministic repeat hashes/tolerances, wall/CPU time, and peak memory. Compile/link qualification precedes solver execution; case-specific equality does not silently generalize to another mesh, thread count, or MPI decomposition.

Direct answer. The limitation is best understood as an implementation thread-safety and inter-rank-state problem, not as “UEXTERNALDB forbids parallelism.” The present code nevertheless remains serial-qualified only, with one small case-specific threaded repeat; refactoring and staged tests are required before parallel production use.

10. Claim matrix: 24 July versus 11 August

Claim	24 July	11 August	Evidence	Remaining limitation
MISESERI pre-refinement	validated in scope	SUPPORTED	offline/native candidate generation	indicator is not fracture solution
Uniform verification	partial Mode-I roles	PASS pre-peak	H2 reproduced its earlier path and revealed an interior peak; jobs 1386372, 1386447, 1388330	H1/H2 diverge before endpoint; peak convergence unresolved
Adaptive global response	pre-peak refined-v3	HOLD	1386469/70 agree mutually ($\sim$ 0.16% maximum RF difference), but late force is about half the uniform level	primary adaptive curves require curve-level reconciliation
Adaptive crack path	not equivalent	FAIL gate	H1/H2 matched Hausdorff 0.005443 mm	gate 0.003750 mm
Nonmatching transfer	bounded offline/ingestion	artifact prepared	MM-to-target offline mapping	active initialization path did not consume it
Mechanical continuation	unproven	NOT DEMONSTRATED	1386471 followed the virgin PK5 trajectory	requires explicit ingestion
Runtime ingestion	bounded older test only	FAIL current path	UEL ignores SVARS/artifact	reader/init redesign
Evolving remesh	withheld	HOLD	second controlled stage is not scientifically executable	dependent on ingestion proof
COMMON thread safety	not general	UNRESOLVED	D2C positive; static risks; P3-T4 compile fail	general equality absent
MPI / hybrid safety	not established	UNRESOLVED	no valid multi-rank/hybrid science	distribution design absent
Online remeshing	withheld	NOT YET CLAIMED	offline CAE remesh/candidate work only	no closed online loop

Resolved or materially advanced. Corrected FRACFIX/N_{phys} formulation; H2 trajectory reproduction and an identified interior peak; scoped pre-peak uniform consistency; two endpoint-complete adaptive calculations; and a precise parallel risk model.

Still unresolved. Complete uniform post-peak reference, crack-path convergence, actual transferred-state restart, second evolving remesh, online loop, and general parallel safety.

11. Limitations and the next 2–4 weeks

Current limitations. H1 and H2 both terminate by numerical divergence before the prescribed endpoint, so global peak and full post-peak convergence remain unresolved. The frozen crack-path gate fails, no complete uniform reference exists, direct validated fracture-energy history is unavailable, and H2 contains eight small SDV15 decreases. Although the adaptive meshes agree closely with each other, their late force differs materially from the uniform references; adaptive-to-uniform accuracy and accuracy-versus-cost therefore remain on hold pending primary-curve reconciliation. Runtime state ingestion is not demonstrated, and the second controlled transfer stage remains on hold. Native/offline mesh creation is not an online loop. General thread, MPI, and hybrid safety remain unresolved.

Scientifically required

1. Implement and prove artifact-to-target runtime initialization for nodal phase and integration-point history.
2. Qualify the corrected initialization path before any newly authorized execution, then verify source → artifact → first solver state.
3. If that passes, perform one controlled second evolving-remesh transfer and matched-state audit.
4. Retrieve the primary adaptive curves and complete the adaptive-to-uniform RF–U/work audit before making an accuracy-versus-cost claim.
5. Complete P-A/P-B and design the thread-safe state boundary before further parallel tests.

Optional sensitivity work

1. Additional corridor refinement only if the matched-state crack metric motivates it.
2. Alternative transfer projection only after runtime ingestion is proven.
3. MPI/hybrid performance study only after equality and determinism gates pass.
4. No further H2 rerun: the extended calculation has answered the outstanding H2 termination question.

Scientific progression gate. Do not execute the second controlled transfer stage or claim evolving/online remeshing until the executable initialization path consumes the state and the first solver state is independently verified.

12. Reproducibility appendix

Item	Frozen identity or location
Evidence snapshot revision	`0b36c8ed3e5420f34ed66f6468c5994ad3f3e0f4`
H2 endpoint closeout revision	`0b36c8ed3e5420f34ed66f6468c5994ad3f3e0f4`
Runtime-ingestion audit revision	`d2c9c6a6d5f8dce94924ecb9cf6ab77fec96d7be`
Parallel-study closure revision	`8d920db09bb79846fe15697d785d37a012426638`
Current production UEL SHA-256	`0bc4378179a35acd9954d20d3e07517f8e1c356ae07a23c40e7715cd7b56dce8`
Uniform jobs	H0 `1386372`; H1 `1386447`; earlier H2 `1386448`; extended H2 `1388330`
Adaptive jobs	MM `1386469` (M2ADAPT_MM_FRACFIX_PROD); PK5 `1386470` (M2ADAPT_PK5_FRACFIX_PROD)
Restart audit	`1386471` (M2STATE_FRACFIX_RESTART1); primary audit: `project_coordination/sessions/2026-08-11_0525_gemini-antigravity_F43STATE-M2-RUNTIME-INGESTION-AUDIT1.md`; revision `d2c9c6a6d5f8dce94924ecb9cf6ab77fec96d7be`
Uniform evidence	`models/generated/mode_ii/reference_convergence/`; Pair-2 censoring audit
Adaptive evidence	`models/generated/mode_ii/production_adaptive_batch/`
Transfer evidence	`models/generated/mode_ii/production_state_transfer_batch/`
Parallel evidence	`runs/hpc/stage_p/`; `results/validation/stage_p_static_audit/`
Software	Abaqus 2023; Intel 2024.2.0 and GCC 11.4.0 environment; Abaqus Python runtime; serial jobs reported here use one CPU

Glossary. H0/H1/H2 are the coarse/intermediate/fine uniform meshes. MM is the lower-cost locally refined mesh; PK5 is the finer 5%-criterion crack-corridor mesh. FRACFIX is the current corrected phase-field UEL formulation. N_{phys} is the number of physical elements, excluding coincident logical/output layers. D2C is the small one-rank/four-thread repeatability case; P3-T4 is the threaded qualification attempt that stopped during compilation.

Evidence notes. Numerical values come from primary scheduler/solver records, extracted curves, frozen comparison contracts, endpoint closeout, and the executable-path audit. H2 has a defensible interior peak; cross-mesh global peak convergence remains unresolved because H1 is incomplete. H0/H2 available-domain RF $L_2 = 1.891\%$ and work difference $= 1.304\%$ through 0.0097625 mm are contextual, not full-path convergence. The report does not expose credentials or reproduce bulky solver artifacts.

Parallelization references. SIMULIA, *About User Subroutines and Utilities*: `https://docs.software.vt.edu/abaqusv2024/English/SIMACAEANLRefMap/simaanl-c-subroutineover.htm`; and *About Parallel Execution*: `https://docs.software.vt.edu/abaqusv2024/English/SIMACAEEXCRefMap/simaexc-c-parallelmodes.htm` (accessed 11 August 2026).

Current claim boundary in one sentence. The work now supports scoped Mode-II pre-peak uniform consistency and mutual agreement of two endpoint-complete adaptive solutions, but adaptive-to-uniform accuracy, converged crack geometry, actual nonmatching runtime state ingestion, a second evolving remesh, online adaptivity, and generally safe parallel execution are not yet demonstrated.